

Thomas Istvan

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EDUCATION

Cornell University, College of Engineering

Ithaca, NY

Master of Engineering in Mechanical Engineering (Robotics & Controls)

Aug 2026 – May 2027

- MEC Fellowship (\$30,000) and Academic Tuition Scholarship (\$6,000)

University of Hawai'i at Mānoa, College of Engineering

Honolulu, HI

Bachelor of Science in Mechanical Engineering, Summa Cum Laude, GPA: 3.98

Aug 2023 – May 2026

Selected Coursework: Robot Perception • Model-Based Estimation • Autonomous Mobile Robots • Manipulator Robotics • Legged Robots • Swarm Robotics • Maritime Robotics • Soft Robotics • Continuum Mechanics

TECHNICAL SKILLS

Software: Python, MATLAB, ROS 2, Gazebo, MoveIt 2, KiCad, C++, MuJoCo, Linux, Git/GitHub, PyTorch

Hardware: SolidWorks, Onshape, COMSOL, FDM & resin 3D printing, CNC/mill/lathe, Arduino, Teensy, Raspberry Pi, I2C/UART/SPI/USB, PWM motor actuation, soldering, and tensile testing

EXPERIENCE

Robotics Engineer, ASML

Ithaca, NY | Sept 2026 – Present

- Re-architecting a legacy LabVIEW control system into a modular ROS 2 stack of custom C++/Python nodes for a reticle-handling robotic arm
- Validating control logic and testing safety in a Gazebo simulation with MoveIt 2 before running on the arm
- Enabling closed-loop, real-time velocity control of a multi-axis arm by deriving forward/inverse kinematics and Jacobian-based force mapping that convert end-effector targets into joint-space commands

Mobile Robotics Research Assistant, RIP Lab

Honolulu, HI | May 2024 – July 2026

- Achieved stable vertical-surface operation for a 50-lb magnetic climbing robot by optimizing wheel and magnet geometry through MATLAB and COMSOL simulation before fabrication
- Cut prototype iteration time by designing parts such as sensor housings and caster wheels in Onshape and producing them in-house with FDM and resin 3D printing
- Provided real-time orientation feedback by building a C++ ROS 2 pipeline that fuses dual 9-DOF IMU streams with an Extended Kalman Filter and visualizes the state live in RViz2
- Built an enclosure with cooling fan, heater, and I2C temperature sensor, with a Python tool to track temperature within 5% of a reference sensor, simulating heat generation from batteries in the enclosure.

Lead Systems Engineer, Team Kanaloa

Honolulu, HI | Aug 2025 – May 2026

- Led a team building a 400-lb autonomous surface vessel; validated the Python/C++ PID control and autonomy stack in Gazebo (URDF model, checked against a Fossen hydrodynamic model) before on-water testing
- Achieved centimeter-level real-time localization by implementing an EKF node that fuses 3 GPS modules and 2 IMUs with RTCM corrections
- Wrote low-level ROS 2 drivers (I2C, UART, SPI, USB), designed a custom sensor PCB in KiCad, and integrated ESCs with live feedback and a Ubiquiti radio link for real-time communication during field testing
- Ran comparative FEA in SolidWorks across hull composite layouts to select a design with a factor of safety of 5 in 4-foot wave conditions

PROJECTS

Unitree Go2 X Quadruped Robot Dog

Ithaca, NY | Aug 2026 – Present

- Building a MuJoCo simulation environment of the Go2 and training a locomotion policy with deep reinforcement learning (PPO) in PyTorch
- Designing the observation/action spaces, reward function, and training curriculum, with domain randomization to improve sim-to-real transfer, then deploying the trained policy and validating the pipeline via unit tests

Autonomous Path Planning of a Differential Drive Robot

Ithaca, NY | Aug 2026 – Present

- Building a ROS 2 autonomy stack with Nav2 and ros2_control in Gazebo that maps unknown environments with SLAM using a lidar, wheel odometry, and slam_toolbox to navigate autonomously
- Implementing global path planning with A* over an inflated occupancy-grid costmap and tracking paths with a DWB local planner for dynamic obstacle avoidance
- Tuning costmap inflation, controller, and recovery-behavior parameters to achieve robust goal-reaching in simulation

LEADERSHIP

Graduate Teaching Assistant, Cornell University

Ithaca, NY | Aug 2026 – Present

- Teach open- and closed-loop feedback control concepts (System Dynamics, MAE 3260) to 21 undergraduates in weekly labs
- Train students on lab equipment such as oscilloscopes, accelerometers, and proximity sensors, and hold office hours for report-writing guidance